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Appendix

The appendix is organized as follows: In Appendix A, we present preliminary experiments comparing SVG and image-based representations. In Appendix B, we include details on error analyses, and in Appendix C, we describe Primal Visual Description details. Appendices D–F present additional experiments on Vtracer visual encoding quality, PVD model variants, and open-source LMM reasoner variants. Appendix G provides additional qualitative examples of PVD parsing novel concepts via composition. Appendix H presents further efforts in refining prompts for integrating PVD into LMMs. Appendix I shows the full input and output from GPT-4 for the maze-solving example depicted in Figure 2. Task prompts and newly constructed downstream task datasets can be found in Appendices J and K, respectively. In Appendix L, we include detailed statistics for all of the datasets we used.

A Preliminary Experiments on SVG Representations

We introduce a suite of probing tasks to evaluate current LMMs’ capabilities in performing tasks with vector graphics. The results show that even state-of-the-art LMMs, such as GPT-4V, struggle with tasks that require precise perception of low-level primitives, such as comparing the lengths of two lines. We then investigate where this deficiency originates and propose an alternative representation, Scalable Vector Graphics (SVG), for representing such precise low-level features. We find that, compared to image-based representations, SVG representations can be more efficient for visual reasoning on vector graphics. However, they are not without their own limitations, which we will elaborate on in § A.3.

A.1 Image and SVG Representations

In the probing tasks, we include both discriminative and generative tasks, each with varying levels of emphasis on low-level visual details. Illustrations of the input and output examples are available in Figure 8. We additionally include a non-vector-graphics task, Clevr QA, which consists of realistic 3D rendered scenes. This is to test the limits of SVG representations in encoding 3D objects within realistic images. Detailed statistics of these tasks can be found in Table 13.

For each task, we consider two evaluation settings: zero-shot and fine-tuning. We explore two types of representations for the input image: (1) direct use of the image pixels, encoding them into patch embeddings with an image encoder, e.g., CLIP (Radford et al., 2021); (2) conversion of the image into SVG code using a rule-based raster-to-SVG converter (VTracer, 2024).

For fine-tuning with the image input, we instruction-finetune Llava-v1.5-7b (including the LLM-backbone and the projection layer) using Lora (Hu et al., 2022) on the training set for one epoch. For fine-tuning with the SVG input, we only fine-tune the LLM backbone of Llava-v1.5, Vicuna (Chiang et al., 2023), using Lora for one epoch, with the input image’s SVG code concatenated in the context. The results are shown in Table 3. Key observations include:

- (1) The SOTA open-source LMM, Llava-v1.5, struggles to achieve non-trivial performance on most probing tasks even with dedicated fine-tuning. On tasks with binary choices, Llava tends to predict homogeneous answers, disregarding differences in the input image.
- (2) The SOTA closed-source LMM, GPT-4V, excels on task Line or Angle, which focuses on querying the high-level semantics of the primitive concept (“what’s in the image”). However, its performance significantly decreases on tasks requiring more precise low-level perception, e.g., Angle Classification and Length Comparison.
- (3) Fine-tuning the LLM backbone, Vicuna, with SVG inputs consistently outperforms fine-tuning the entire Llava model with image inputs. This highlights the potential of using SVG as an alternative representation in vector graphics.
- (4) We note that SVG may inherently be inefficient in representing rendered 3D scenes and realistic images due to factors like camera perspectives, lighting, and shadows. While our focus in this work is on vector graphics, we leave the extension to other domains for future exploration.